VIRTUAL UNIVERSITY OF PAKISTAN

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(Final Term Past Paper)

MADE AND SOLVED BY TEAM HADI

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Best of luck!

PAST PAPER

Question No : 1 of 52 | Marks: 1 (Budgeted Time 1 Min)

In integrating $\int_0^{\frac{\pi}{2}} \sin x dx$, by dividing the interval into four equal parts, width of the interval should be

Answer (Please select your correct option)

- π
- $\frac{\pi}{2}$
- $\frac{\pi}{4}$
- $\frac{\pi}{8}$

Find the approximate value of

$y \sin x dx \; \pi = \int_0$ using

(i) Trapezoidal rule

(ii) Simpson's 1/3 rule by dividing the range of integration into six equal parts. Calculate the percentage error from its true value in both the cases.

Solution

Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

The minimum interval in which the root of the equation $x^2 - 3x + 1 = 0$ lie is

Answer (Please select your correct option)

- (0, 1) ✓
- (1, 2)
- (2, 4)
- (0, 2)

Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

If the root of the given equation lies between a and b, then the first approximation to the root of the equation by bisection method is ……

Answer (Please select your correct option)

- $\frac{(a+b)}{2}$ ✓
- $\frac{(a-b)}{2}$
- $\frac{(b-a)}{2}$
- None of the given choices

Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

Next approximation to the root of the given equation by bisection method can be found if

because no given value !

Answer (Please select your correct option)

- $f(2) = -7,\ f(3) = -8$
- $f(2) = -7,\ f(3) = 8$
- $f(2) = 7,\ f(3) = 8$
- None of the given choices ✓

Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

If $8+7i$ is the root of a quadratic equation $f(x)=0$, then the other root of the given equation will be

root can b real or imag

Answer (Please select your correct option)

- $8-7i$ ✓ not sure
- $-8-7i$
- $-8+7i$
- 8

Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

For the equation $x^3 + 3x - 1 = 0$, the root of the equation lies in the interval......

Answer (Please select your correct option)

- (1, 2)
- (1, 3)
- (0, 1) ✓
- (1, 2)

Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

The quantity of error which is present in the statement of the problem itself ,before finding its solution is called

Answer (Please select your correct option)

- Inherent error
- Local round off error
- Local Truncation error
- Typing error

Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

If there are two roots of the equation and one root is $2+3i$, then the other root will be

Answer (Please select your correct option)

- $2+3i$
- $2-3i$
- $3+2i$
- $3-2i$

Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

It can be verified that for any matrix [A]

Answer (Please select your correct option)

- $[A][A^{-1}]=[I]$, I = identity matrix
- $[A][A^{-1}]=[D]$, D = diagonal matrix
- $[A][A^{-1}]=[S]$, S = symmetric matrix ✓
- $[A][A^{-1}]=[Z]$, Z = orthogonal matrix

Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

In Crout's reduction method the elements of the main diagonal of upper triangular matrix are taken as

book

Answer (Please select your correct option)

- 0
- 1 ✓
- -1
- None of the given choices

None of the given choices

Crout's ReductionMethod
Here the coefficient matrix [A] of the system of equations is decomposed into the product of two matrices [L] and [U], where [L] is a lower-triangular matrix and [U] is an upper-triangular matrix with 1's on its main diagonal.

Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

In Gauss-Seidal method, the new approximation calculated is

in Gauss-seidel method the new approximation calculated is instantly replaced by the previous one.

Answer (Please select your correct option)

- Used in the next iteration for next approximation
- Instantly replaced by the previous one
- Never used again
- None of the given choices

Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

In relaxation method, for fast convergence all the terms should be taken to one side and then reordering should be done so that the largest coefficients should appear on the

To achieve the fast convergence of the procedure, we take all terms to one side and the reorder the equations so that the largest negative coefficients in the equations appear on the diagonal.

Answer (Please select your correct option)

- End of the rows
- Beginning of the rows
- Diagonal
- None of the given choices

Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Every square ------- matrix has an inverse.

Every square non-singular matrix will have an inverse.

book

Answer (Please select your correct option)

- Singular
- Non-singular
- Zero
- None of the above

Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

If S= $\begin{pmatrix} 1 & 2 & 3 \\ 6 & 4 & 7 \\ 8 & 6 & 5 \end{pmatrix}$, then

Answer (Please select your correct option)

- $S^{-1} = S^T$ ✓ not sure
- $S^T = S$
- $I = S$
- None of given choices

Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

A 3×3 matrix $[A]$ is said to be orthogonal if

Answer (Please select your correct option)

- $[A]^{-1}[A]=[I]$
- $[A]^{T}[A]=[I]$
- $[A]^{T}[A]=[O]$
- None of the given choices

Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Jacobi's method for finding the eigen values can be applied to

Answer (Please select your correct option)

- Real and symmetric matrix
- Real and unsymmetric matrix
- Real and complex unsymmetric matrix.
- None of given choices.

Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

Lagrange's interpolation formula is used when the values of the independent variable are

Answer (Please select your correct option)

- Equally spaced
- Not equally spaced
- Constant
- None of the above

Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

If there are (n+1) values of y corresponding to (n+1) values of x, then we can represent the function f(x) by a polynomial of degree

Answer (Please select your correct option)

- n+2
- n+1
- n
- n-1

 Marks: 1 (Budgeted Time 1 Min)

Given the following data

x	1	3	7
$f(x)$	2	4	10

Answer (Please select your correct option)

- Newton's forward difference interpolation formula
- Newton's backward difference interpolation formula ✓
- Lagrange's interpolation formula
- None of the given choices

Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the following data

x	0	1	4
$y = f(x)$	2	1	4

Answer (Please select your correct option)

- -1

incomplete

- 1 ✓

not sure

- 2
- 4

Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Relationship between differential operator and backward difference operator is given by

Answer (Please select your correct option)

- $hD = -\log(1+\nabla)$
- $hD = \log(1+\nabla)$
- $hD = \log(1-\nabla)$
- $hD = -\log(1-\nabla)$

Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Shift operator is defined as

Answer (Please select your correct option)

- $Ef(x) = f(x-h)$
- $Ef(x) = -f(x-h)$
- $Ef(x) = -f(x+h)$
- $Ef(x) = f(x+h)$

Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given table of values

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

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using two-point equation the value of $f'(0.5)$ is.........:

Answer (Please select your correct option)

- 0.73
- -0.75
- 0.5 ✓
- -0.5

Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given table of values

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

Answer (Please select your correct option)

- 17.5
- 12.5
- 7.5
- -2.3

formula for fining this type of questions

f(x1)=x2-x1/h

incomplete

Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

The magnitude of truncation error in Milne's predictor formula is

Answer (Please select your correct option)

- $\frac{1}{90}h\Delta^4 y_0'$ ✓
- $\frac{1}{90}h\Delta^3 y_0'$
- $\frac{28}{90}h\Delta^3 y_0'$
- $\frac{28}{90}h\Delta^4 y_0'$

Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Let I denotes the closed interval spanned by $x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7, \overline{x}$. Then $F(x)$ vanishes ------times in the interval I.

Answer (Please select your correct option)

- 6
- 9
- 7 ✓
- 8

Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

From the following table of values:

x	1.3	1.5	1.7	1.9	2.1	2.3
y	2.9648	2.6599	2.3333	1.9922	1.6442	1.2969

Estimation of $y'(1.3)$ will be done using the formula of differential operator in terms of

Answer (Please select your correct option)

- Backward difference operator ✓
- Central difference operator
- None of the given choices
- Forward difference operator

Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

While deriving Trapezoidal rule, we approximate f(x) in the form ....

Answer (Please select your correct option)

- $ax+b$ ✓
- ax^2+bx+c
- ax^3+bx^2+cx+d
- $ax^4+bx^3+cx^2+dx+e$

Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

While deriving Simpson's 3/8 rule, we approximate f(x) in the form ………

Answer (Please select your correct option)

- $ax + b$
- $ax^2 + bx + c$ ✓
- $ax^3 + bx^2 + cx + d$
- $ax^4 + bx^3 + cx^2 + dx + e$

Question No : 30 of 52

Marks: 1 (Budgeted Time 1 Min)

Two segment trapezoidal rule of integration is exact for integrating at most .... order polynomial.

Answer (Please select your correct option)

- first
- second
- third ✓
- fourth

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

To apply Simpson's 3/8 rule, the number of intervals can be....

Answer (Please select your correct option)

- 4
- 7
- 9
- 8

bht confuse ha past paper may bht sy
answer ha i think 7,8 may sy koyi

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

To apply Simpson's 3/8 rule, the number of intervals can be....

Answer (Please select your correct option)

- 4
- 7
- 9
- 8

Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

Romberg's integration method is ---------- than Trapezoidal and Simpson's rule.

Answer (Please select your correct option)

- none of the given choices
- more accurate ✓
- less accurate
- equally accurate

Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The global error in Simpson's 3/8 rule is of

Answer (Please select your correct option)

- $o(h^2)$ ✓
- $o(h^3)$
- $o(h^4)$
- None of these choices

Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Rate of change of any quantity with respect to another can be modeled by

Answer (Please select your correct option)

- An ordinary differential equation
- A partial differential equation
- A polynomial equation
- None of the given choices

Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

A third order ordinary differential equation can be reduced to a system of ---------- first order ordinary differential equations.

Answer (Please select your correct option)

- 0
- 1 ✓
- 2
- 3

Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Given $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition y=1.02 at t=0.02. Using Euler's method, y at t= 0.04, h=0.02 is

Answer (Please select your correct option)

- 3.0392
- 2.0392
- 1.0392
- 0.0392

Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

In fourth order Runge-Kutta method, k_1 is given by

Answer (Please select your correct option)

- $k_1 = hf(x_n, y_n)$ ✓
- $k_1 = 2hf(x_n, y_n)$
- $k_1 = 3hf(x_n, y_n)$
- None of the given choices

Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

In fourth order Runge-Kutta method, k_4 is given by

Answer (Please select your correct option)

- $k_4 = hf(x_n + 2h, y_n + 2k_3)$
- $k_4 = hf(x_n - h, y_n - k_3)$
- $k_4 = hf(x_n + h, y_n + k_3)$ ✓
- None of the given choices

Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

The truncation error in Adam's predictor formula is

Answer (Please select your correct option)

- $\frac{251}{720} h\nabla^4 y_n'$ ✓
- $\frac{251}{720} h\nabla^4 y_{n+1}'$
- $\frac{19}{720} h\nabla^4 y_{n+1}'$
- $\frac{19}{720} h\nabla^4 y_n'$

Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

In solving the differential equation

$$y' = x^2 + 2y \; ; \; y(1) = 3$$

$h = 1$, By Euler's method $y(2)$ is calculated as

Answer (Please select your correct option)

- 4
- 6
- 8
- 10

Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Obtain numerically the solution of

$$y' = x + y^2, \quad y(0) = 2$$

Using Euler's method to find y at x=1, h=1

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

If $f(0) = 3$ and $f(1) = 9$, then find the next approximate value of the function using secant method

Answer (Please click here to Add Answer)

Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Write a backward difference formula of $D^2 f(x)$.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 44 of 52

Marks: 2 (Budgeted Time 4 Min)

Write a formula for Simpson's 1/3 rule.

Answer (Please click here to Add Answer)

Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Evaluate the integral

$$\int_{3}^{5} (\log x + 2)\,dx$$

Using Simpson's 3/8 rule

Take h=1

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Write the two steps of solving the linear equations using Gaussian Elimination method

Answer (Please click here to Add Answer)

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Runge-Kutta Method of order four to find the values of k_1 and k_2 for the initial value problem

$$y' = yx^2, y(0) = 1, h = 0.1$$

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 48 of 52 — Marks: 3 (Budgeted Time 6 Min)

Evaluate the integral

$$\int_0^3 (x^2 + x)dx$$

Using Trapezoidal rule
Take h=1

Answer (Please click here to Add Answer)

100%

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Evaluate the integral

$$\int_0^3 (x+1)dx$$

Using Simpson's 3/8 rule

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Find the dominant eigenvalue and the corresponding eigenvector of the matrix

$$\begin{bmatrix} 8 & 1 & 2 \\ 0 & 10 & -1 \\ 6 & 2 & 15 \end{bmatrix}$$

by Power Method with vector $V^{(0)}=(1,1,1)^T$ as the initial vector.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

From the following table of values, construct backward difference table.

x	0.1	0.2	0.3	0.4
$f(x)$	1.10517	1.22140	1.34986	1.49182

Answer (Please click here to Add Answer)

Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Evaluate the integral

$$\int_0^4 (x^2+1)dx$$

Using Simpson's 1/3 rule

Take h=1

Answer (Please click here to Add Answer)

100%

Normal Arial 12